Natural Language Processing

Instructor

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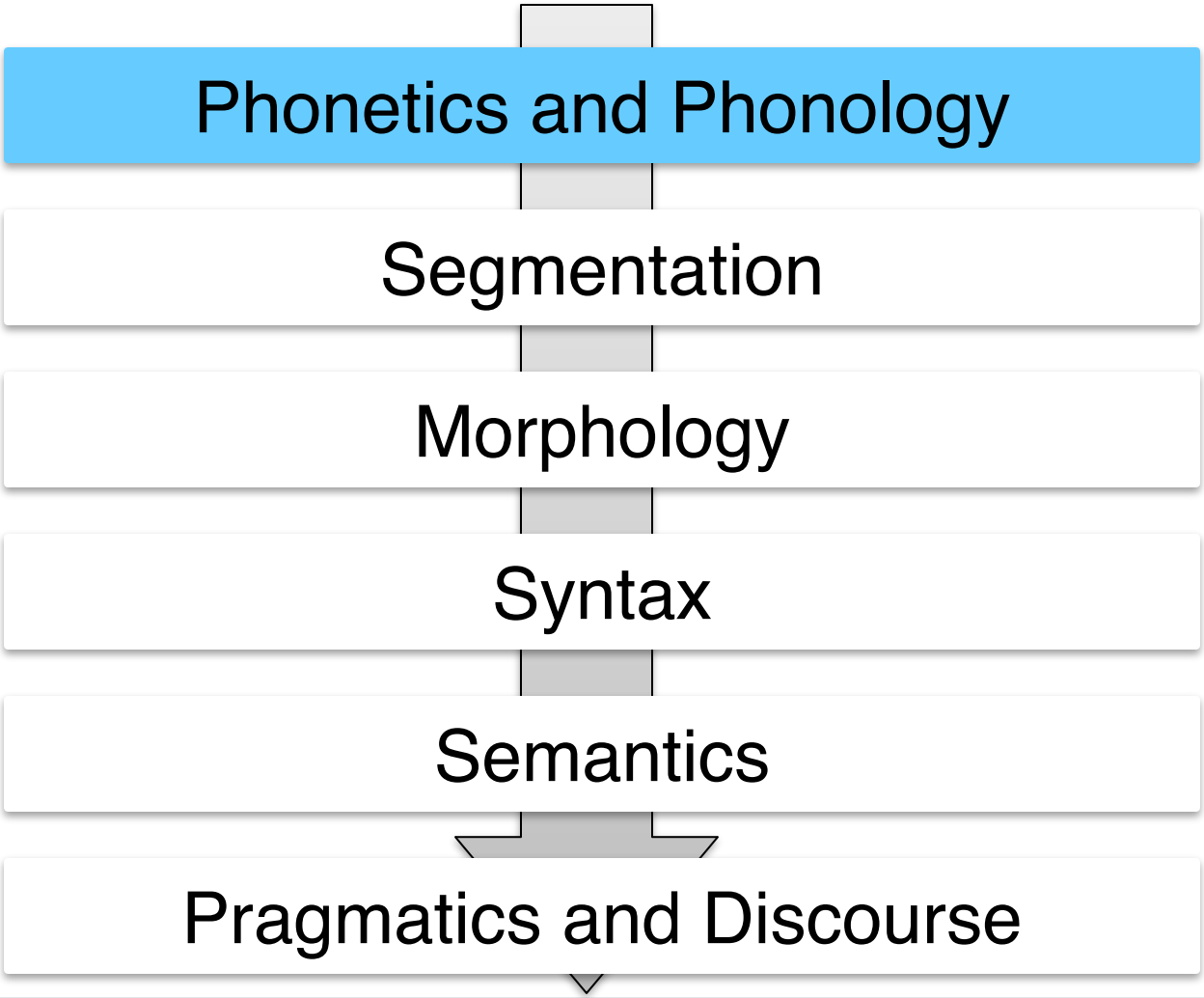
NLP—Analysis Levels



Why Sound?

Why Sound?

- Natural language evolved from sounds, written scripts came later
- Primary step towards knowing a language is to know the sound of language

What is Phonetics?

- Aims to describe all the sounds of all natural languages

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  - Some letters are silent
    - autum**n**, **p**sychology, **k**nife

Phonetic Alphabet

- **International Phonetic Alphabet (IPA)** (1888)
  - One to one mapping to each sound to a phonetic symbol
  - A person, aware of IPA , will be able to pronounce any word in any language
  - Website- https://www.internationalphoneticassociation.org/

Vowel

ʌ	CUP, LUCK
ɑ:	ARM, FATHER
æ	CAT, BLACK
e	MET, BED
ə	AWAY, CINEMA

Consonant

t	TEA, GETTING
tʃ	CHECK, CHURCH
θ	THINK, BOTH
ð	THIS, MOTHER
v	VOICE, FIVE

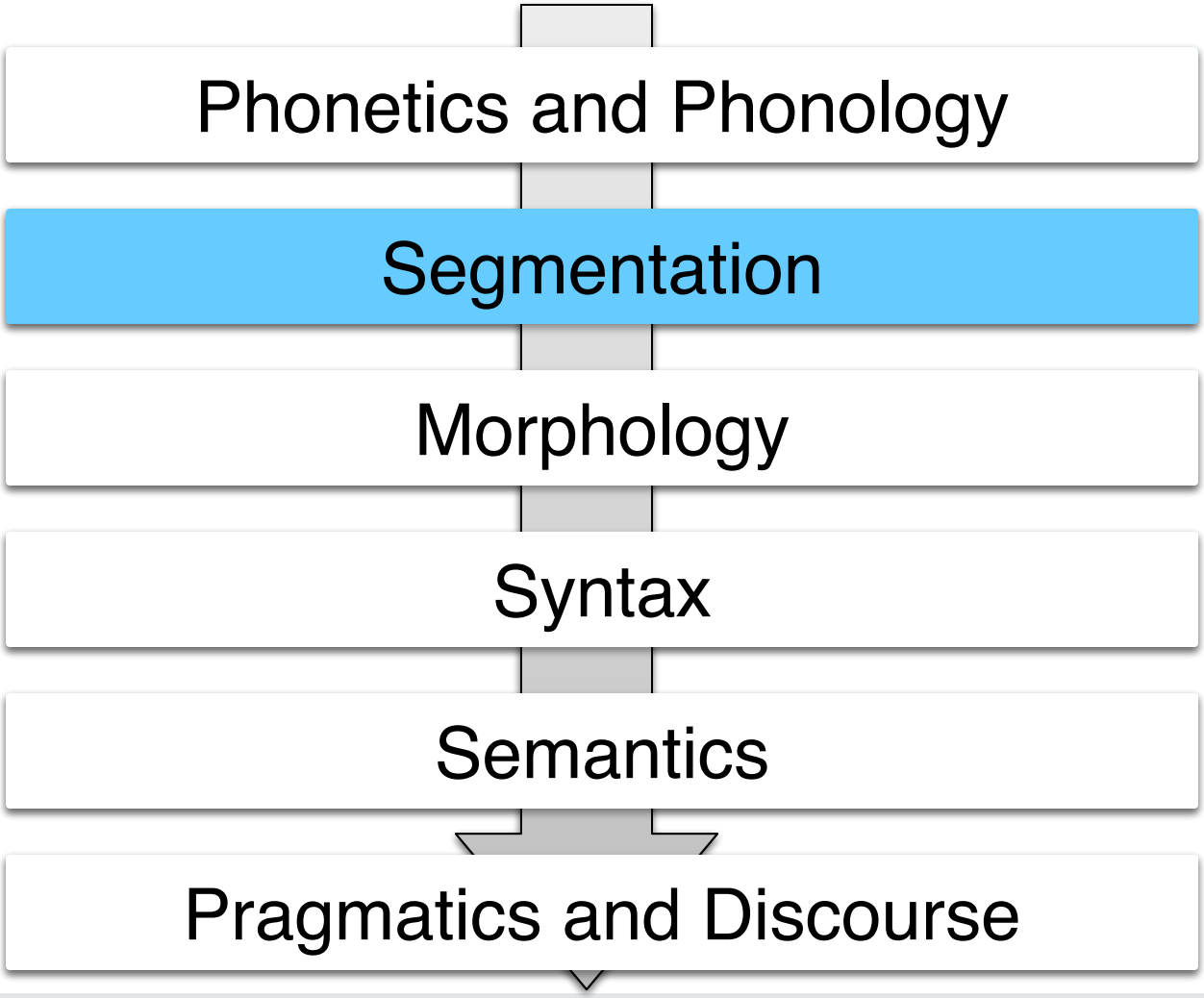
Phonology - Are there patterns of sounds?

- **Phonology** tells us, how various sounds build patterns to form language units

Phonology- Features of speech

- Stress : Making a segment of speech more prominent
- Intonation/Pitch : Describes whether a sound is high or low in volume
- Rhythm : The pattern of putting stress
- Tone: Describes the quality of the sound produced (e.g., sarcastic, friendly, serious,....)
- Syllable structure
  - Syllable is a group of one or more sounds.
  - Essential part of a syllable is a vowel sound (V) which may be preceded and/ or followed by a consonant (C) or a cluster of consonants (CC or CCC)
  - Examples
    - vowel preceded by one consonant (CV) as - *pie* - /*pai*/
    - vowel preceded by two consonants (CCV) as in *try* - /*trai*/
    - just one vowel sound (V) as in *I* and *eye* - /*ai*/, *owe* - /ə/

Analysis Levels in Language Understanding



Segmentation- Why required?

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Text segmentation is the process of dividing written text into meaningful units, such as words, sentences, or topics. The term applies both to mental processes used by humans when reading text and to artificial processes implemented in computers, which are the subject of natural language processing

Segmentation- What is it?

- Human can not assess a stream of text/sound without understanding the inherent concepts/entities/topics

- **Segmentation**
  - divide written text into meaningful units,
    - such as words, sentences, paragraphs.

What is Tokenization?

- Segmenting an input stream into an ordered sequence of tokens is called **tokenization**.

- A system which splits texts into word tokens is called a **tokenizer**

A very simple example:

- Input text:
  John likes Mary and Mary likes John.
- Tokens:
  {"John", "likes", "Mary", "and", "Mary", "likes", "John", "."}

Tokenization- Ambiguities?

Period (.)

- In most cases: The final sentence punctuation symbol
- Part of an abbreviation, e.g. F.D.P.
- Numbers with fractions, e.g. 1.543
- References to resources locators, e.g.: www.apple.com
- To complicate things, if a sentence ends with an abbreviation which ends with a period, only one period is written. "He lives at Lakeview Dr."
- …

Whitespace character

- Part of numbers, e.g. "1 543"
- No segmentation character in multi-word expressions
  - "New York"

Tokenization- Ambiguities?

Comma

- Part of numbers, e.g. 1,543

Single quote

- Within tokens to mark contractions and elisions,
  - e.g. English: *don't, won't, you've, Jame' s new hat*;

Dash

- A delimiter, if it connects strings of digits, e.g. "see page 100-101"
- In **most cases**, however, it is part of the token, e.g. *multi-word*

Tokenization- Other languages?

Chinese 爱国人

- No spaces
- Two possible segmentations, both of them are syntactically and semantically correct
- Disambiguation can only be done with contextual information

爱国 / 人

country-loving person

爱 / 国人

love country-person

Tokenization- General Approaches?

- Based on white space
- Based on regular expressions or rules
- Based on machine learning
  - Binary classifiers that decide whether a certain punctuation is part of a word or not

Sentence Segmentation

- Relatively unambiguous End of sentence punctuation
  - !, ?
- Ambiguity for ‘.’
  - Inc. / Dr. / Mr.
- General approach
  - Binary classifier
    - Looks at a ‘.’
    - End of sentence vs Not end of sentence
    - Hand written rules or machine learning

Tokenization- Tools in practice?

- Spacy : https://spacy.io/
- OpenNLP: https://opennlp.apache.org/
- Stanford CoreNLP: https://stanfordnlp.github.io/CoreNLP/
- deeplearning4j tokenizer: https://deeplearning4j.org/tokenization
- Neural tokenizer: https://github.com/Kyubyong/neural_tokenizer
- Sentence Segmenter, Tokenizer: https://pypi.org/project/segtok/

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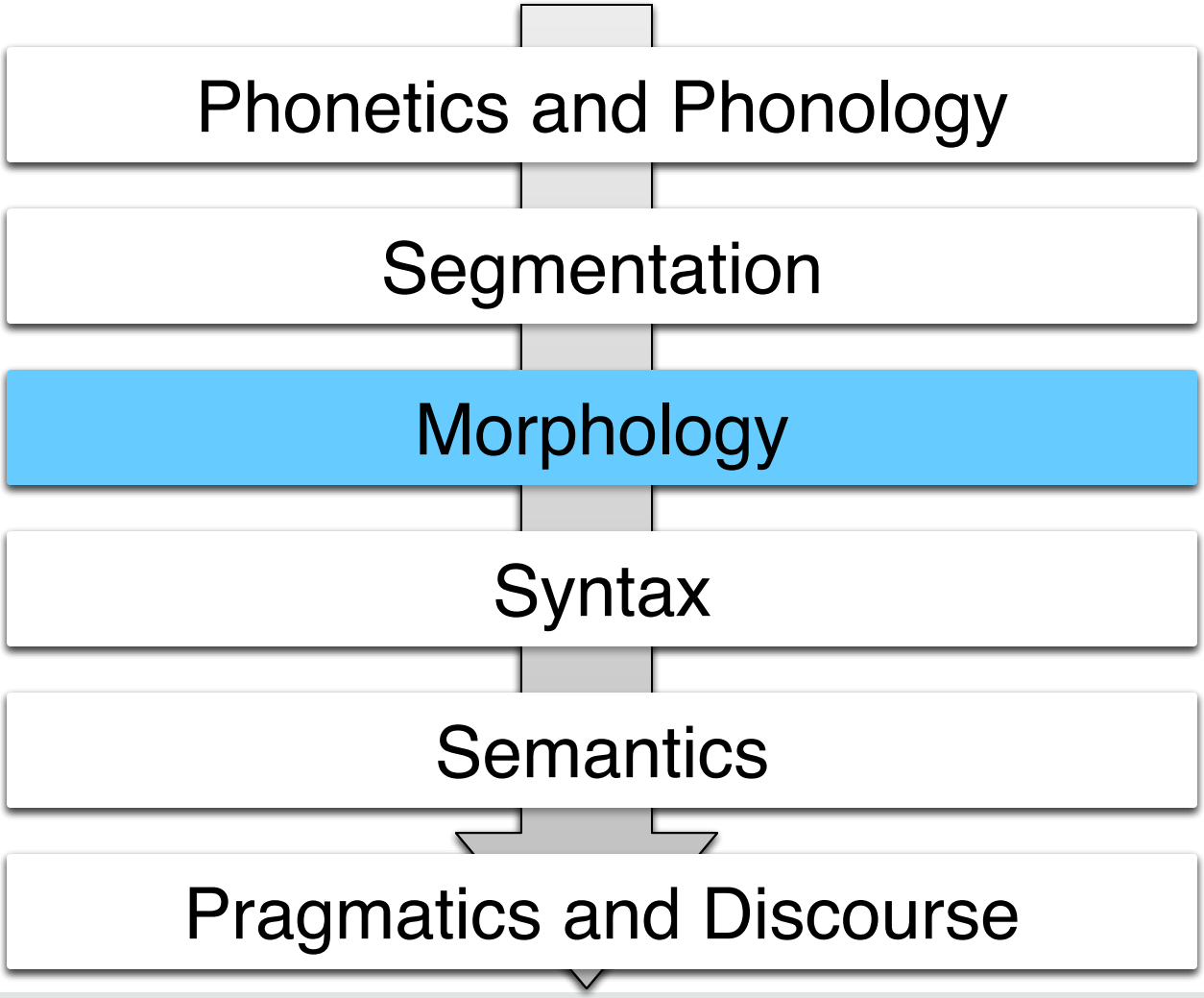

Morphology- Why required?

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lung disease caused by the inhalation of very fine silica dust found in volcanoes

Morphology- Why required?

Words can be further decomposed into smaller units

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What is Morphology ?

- Morphology studies word forms and word formation
- Words are composed of **morphemes**
- Morphemes are the **smallest meaning-bearing units**

- Examples:
  - cats → cat (noun) + s (plural)
  - unknowingly → un + know + ing + ly

Morphemes- Bases and Affixes

- Both *cat* and *cats* can be uttered in isolation but *s* cannot be:
  -*s* is a **bound** morpheme

- Minimal **free** morphemes = **bases** (also called **stems**)
  - *cat* is a free morpheme

- Stems carry the main meaning of the word

- **Affixes** are bound morphemes

Types of Affixes

Suffixes: appear after the base

- cat + s, nice + ly

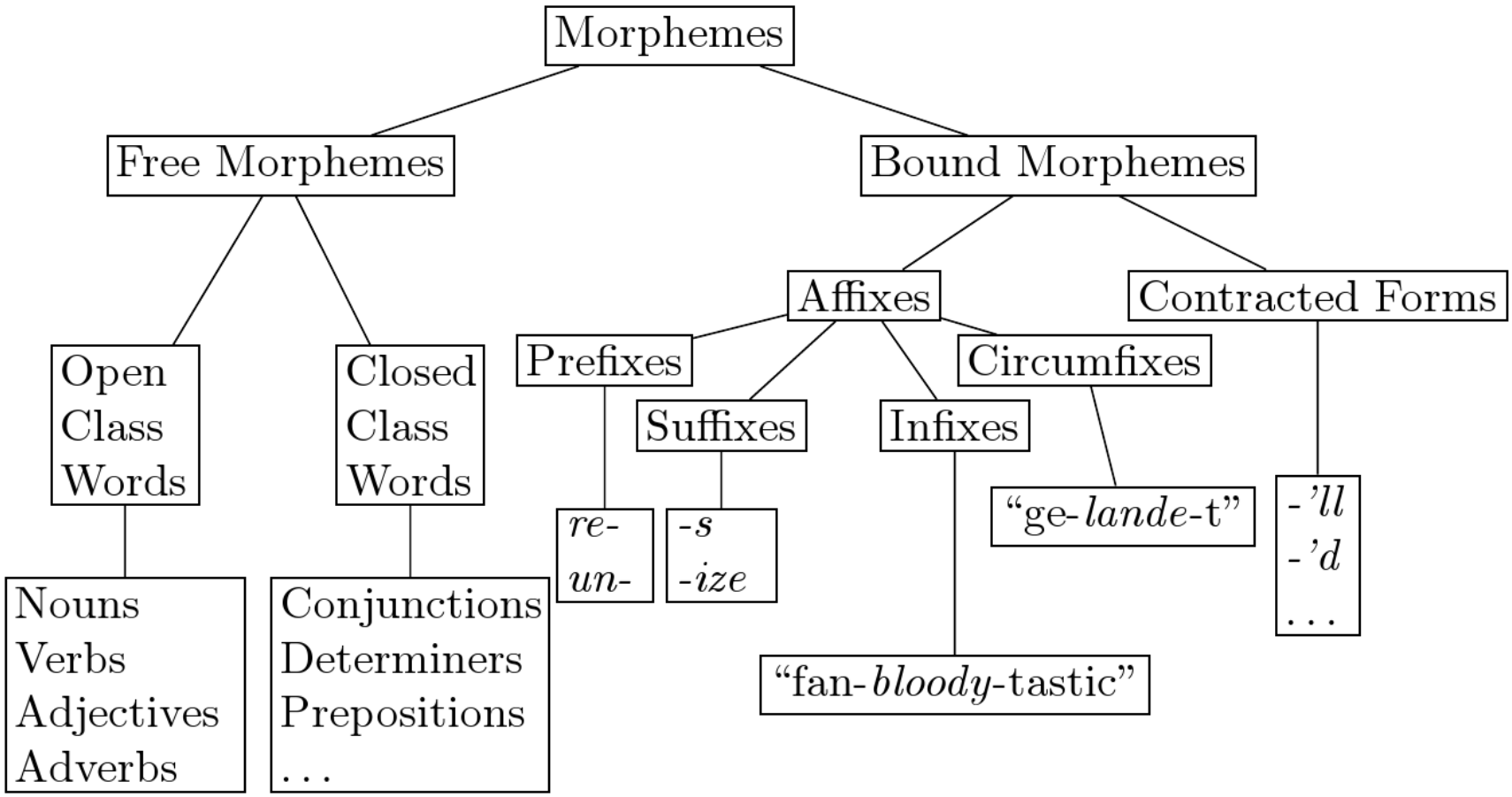
Prefixes: appear before the base

- un + true

Infixes: appear inside the base

- fan + bloody + tastic, un + freaking + believable

Circumfixes: appear on both sides of the base

- en + light + en

Classification of Morphemes